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Practical Work 1: The JavaSTICS interface

We are going to create a simulation with the STICS model. We want to simulate a corn at Mons in the Picardie region in Northern France. The simulation begins in end of winter, the 1st march 2020 (period during which the soil is at field capacity, julian day 61).

I. Create a climate file and the associated weather station file

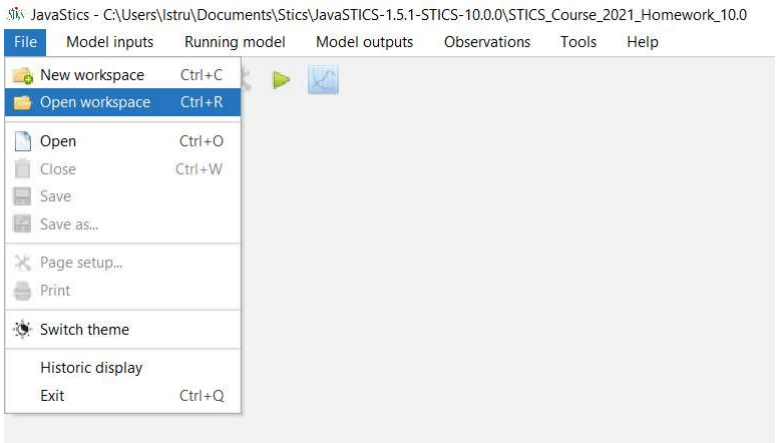
We have climatic data for the year 2020 measured near the experimental site by a weather station of INRAE. Climatic data are from the INRAE database Climatik®. Mandatory variables are minimum temperature, maximum temperature, global radiation and rainfall. Optional variables are PET, wind and vapor pressure.

1. Open the file clim_mons_2020.xlsx which is stored in the working directory Practical_Work_1

2. Replace colon by points
Edition → Replace → , by . → Replace all or use (ctrl + f)

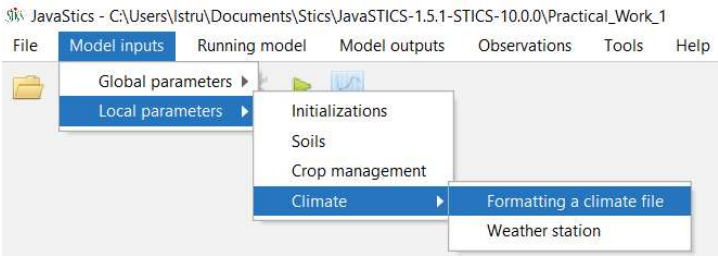
3. Save the file in csv format:
File → Save as → CSV file (separator: semicolon)

4. In STICS, launch JavaStics.exe.
5. Open the file tab and click on "open workspace"



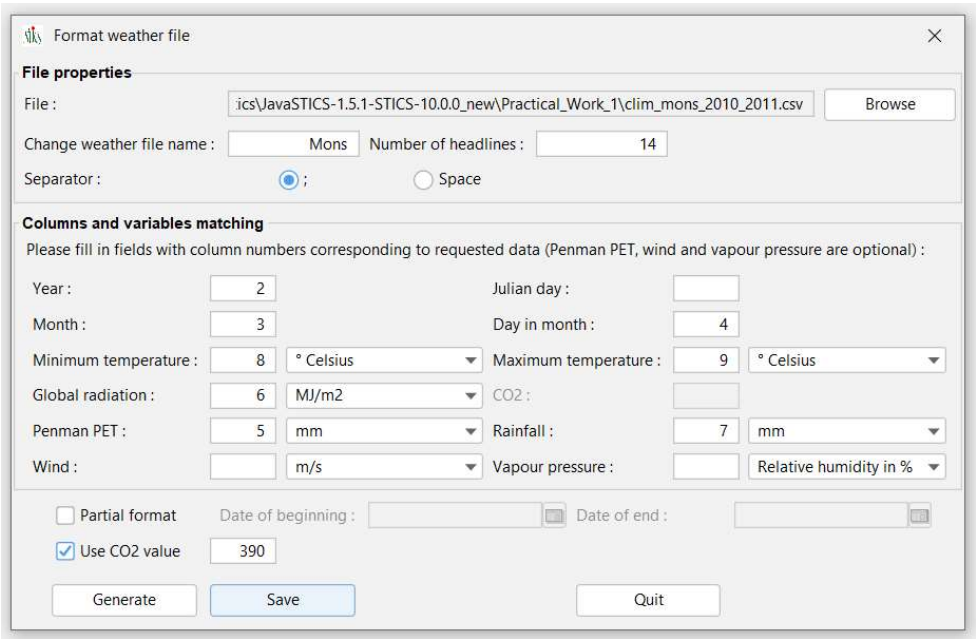
6. Select the working directory Practical_Work_1
7. How do I create a weather file?

/!\ Other ways possible, but the following one is convenient especially if we have a lot of climatic years.



- ✓ Give the pathway to the climate file
- ✓ Give a name to the weather file that will be created = Mons (/!\ has to be the same as the station name)
- ✓ Number of header rows = 14 → File information + name of variables
- ✓ Put the column number that corresponds for each input variable of the model, do not put anything if you do not have the variable.
- ✓ Select the correct units for each variable:

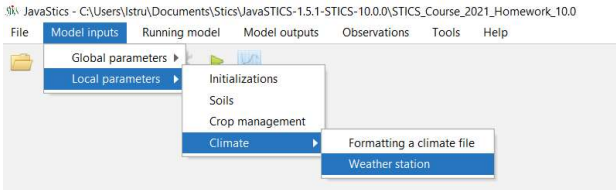
  Temperature (TN and TX) = °C
  Penman PET (ETPP) and precipitation (RR) = mm
  Radiation (RG) = megajoules m^{-2}
  Add a value for CO_2 = 410 ppm (possible to get CO_2 concentration values at: https://gml.noaa.gov/ccgg/trends/)



- ✓ Save
- ✓ Generate

8. Go to the working directory PW1. The weather files have been created = Mons.2010 and Mons.2011

9. How do I create a weather station file?

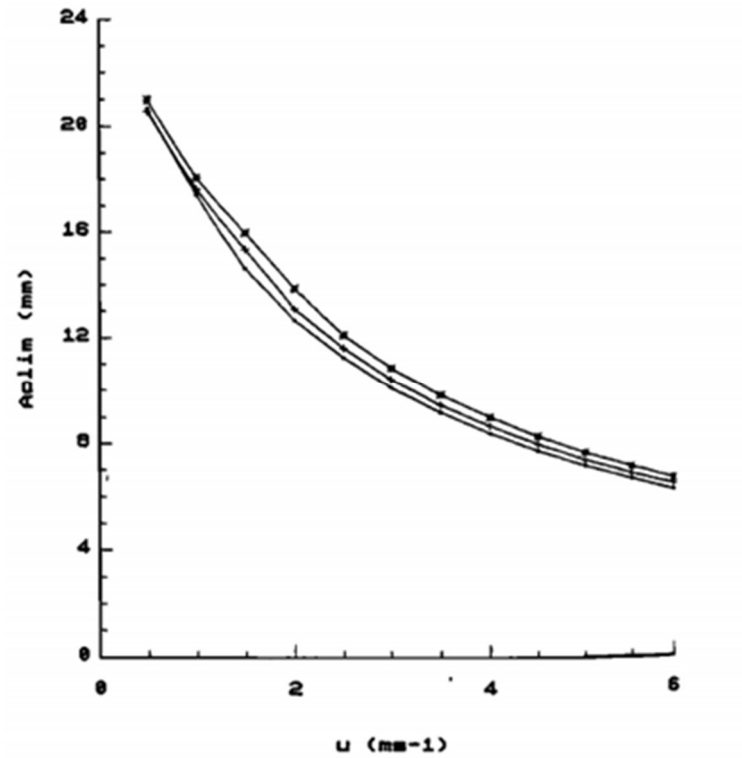


10. Click on "newfile" and give a name to the station file = Mons (/!\ same name as used for the climatic file)

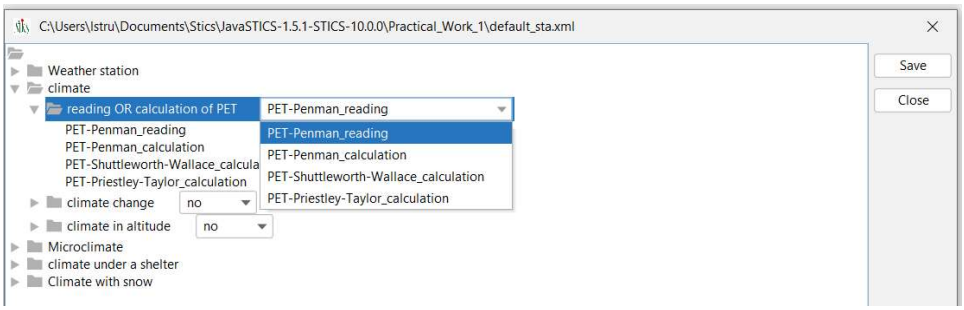
11. Open the file inputs_stics_PW1.xlsx at tab Station:
    - ✓ Open your station file and modify the latitude = 49°

- ✓ Note that mineral nitrogen content input through rainfall can be considered: parameter concRR (keep the default value 0.02 in our case)

- ✓ Set the value of the parameter aclim to 11.8 (function of average wind velocity the days without rain).



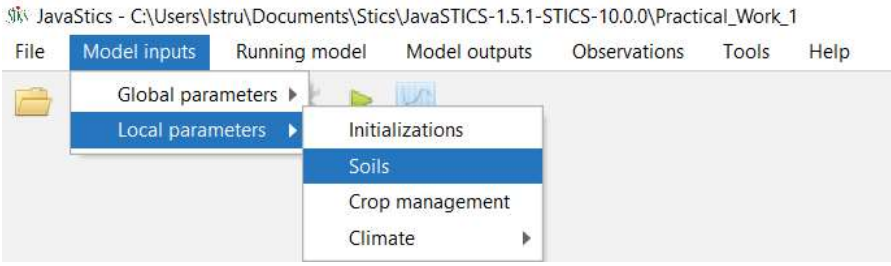
- ✓ Choice of PET calculation in the "climate" tab: depends on the weather data available!



- ✓ **Penman reading**: this is our case, we have this data as an input variable
- ✓ **Penman calculate**: in this case two additional weather variables are mandatory (wind speed and vapour pressure)
- ✓ **Shutwall and Wallace**: in this case two additional weather variables are mandatory (wind speed and vapour pressure). This method of PET calculation is **mandatory for row crops and intercrop.**
- ✓ **Prestley-Taylor**: if you don't have the ETP or wind speed and vapour pressure deficit

II. Create a soil file

1. In JavaSTICS:



2. Create a new soil (Add) → name = Mons.

3. Fill the parameters value thanks to the parameters value given in the tab Soils of the inputs_stics_PW1 excel file

4. In STICS you can have 5 soil layers at most, and this is important to think about what layers you will describe (depend on the soil description, available measurements, kind of processes you are most interested in, …)

Some help to estimate albedo, Q0, epd is available below:

For parameter Q0:

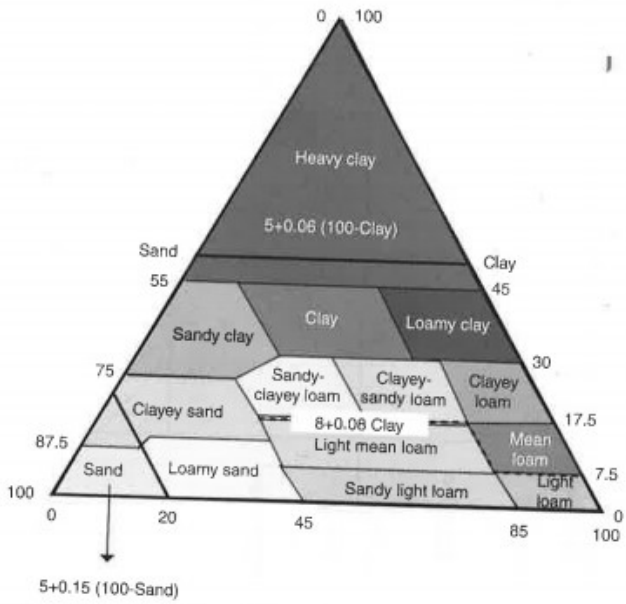


Figure 11.6. Textural triangle and classification by Jamagne *et al.* (1977). Soil transfer functions to assess the $Q0_s$ parameter as a function of clay or sand content.

Type de sol	q0	source
argi < 5	6	Ritchie, 1972
5< argi < 25	9	Ritchie, 1972
25< argi <45	12	Ritchie, 1972
argi > 45	6	Ritchie, 1972
argi = 27	22	Brisson et Perrier, 1991
argi = 51	0	Brisson et Perrier, 1991

For albedo:

tabl. 6 : Values of the dry soil albedo ($ALBEDO_S$) using either textural classes or colours, based on Richard and Cellier (1998), Jacquemoud and Baret (1992) and http://www.icrisat.org/gt-aes/oneds/DataNeeds.htm

Soil type	$ALBEDO_S$
TEXTURE	
Limestone	0.31
Loamy sand	0.25
Clayey loam	0.18 – 0.22
Mean loam	0.22 – 0.23
Crusted mean loam	0.28
Clay	0.28
COLOUR	
Brown soil	0.27
Red soil	0.29
Black soil	0.17
Grey soil	0.29

For epd parameter which controls nitrate dispersion can be estimated from soil textural class and the αpred value in the table below: epd = 2 αpred (be careful, should be in cm in STICS!)

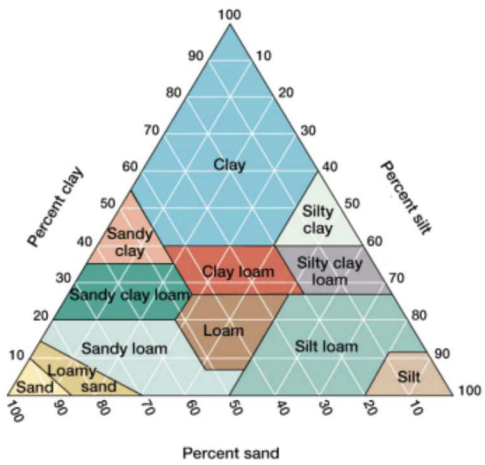


Table 6. Dispersivity (predicted using the regression coefficients in Table 5) as a function of soil textural class.†

Textural Class	n	ψ_a	b	α_{pred}
		kPa		10^{-3} m
Sand	14	0.69	2.79	8
Loamy sand	30	0.36	4.26	26
Sandy loam	124	1.41	4.74	34
Loam	103	3.55	5.25	46
Silt loam	394	7.59	5.33	56
Sandy clay loam	104	1.35	6.77	60
Clay loam	147	2.63	8.17	81
Silty clay loam	324	6.17	8.72	96
Sandy clay	16	0.98	10.73	109
Silty clay	43	3.24	10.39	110
Clay	148	4.68	11.55	128

† Mean values of ψ_a and b from Cosby et al. (1984).

5. SAVE !

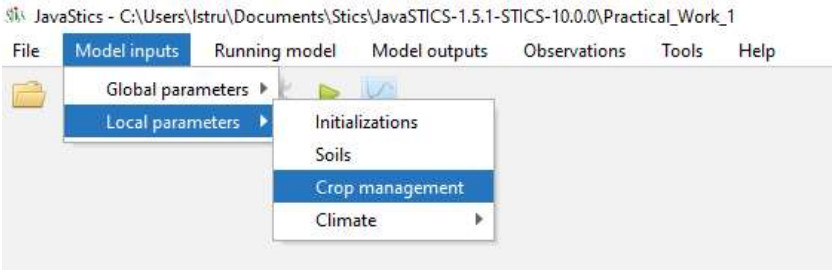
III. Create a technical file

1. We have retrieved the data of the crop management, and here are presented the information we have:

Residues of previous crop: Unknown
Tillage:
05/13/2020 (julian day 134) on 7 cm
06/17/2020 (julian day 169) on 4 cm
Description of crop sowing:
Corn variety DK250 was sown the 5th May 2020 at 4 cm depth with a crop density of 12 plants m^{-2}
Fertilisation:
No
Irrigation:
No
Harvest:
Crop was harvested the 22nd October 2020 and straw were returned to the soil.

This information is summarized in the tab Tec of the inputs_stics_PW1 excel file.

2. In JavaSTICS:



3. Create a new file with the name corn created from the default technical file
/!\ always check the options to clean the technical file if needed!

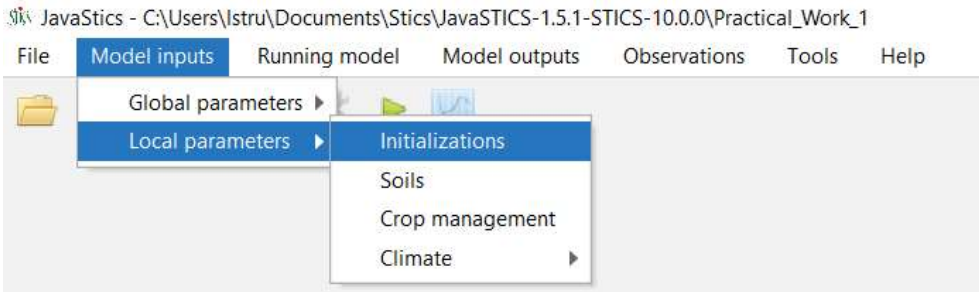
4. Fill the file with the information given earlier (/!\ you can use the tables given in the appendices of the JavaSTICS documentation to transform the given dates in julian days or use an excel formulae)

5. SAVE

IV. Create an initialization file

1. What can I do if I don't have measurements of water and nitrogen to initialize the simulation?
   - ✓ I have information about previous crops and management practices:
     I can launch chained simulations for a minimum of 2 years with a default initialization at the beginning and use model output data on water and nitrogen to initialize the model
   - ✓ I have information about the previous crop but not for management practices:
     Use a default initialization with average data of soil water and nitrogen content observed after cultivation of the previous crop
   - ✓ I have information about soil water and N content but not at the beginning of the simulation:
     You can do a calibration of your initial soil water and nitrogen content

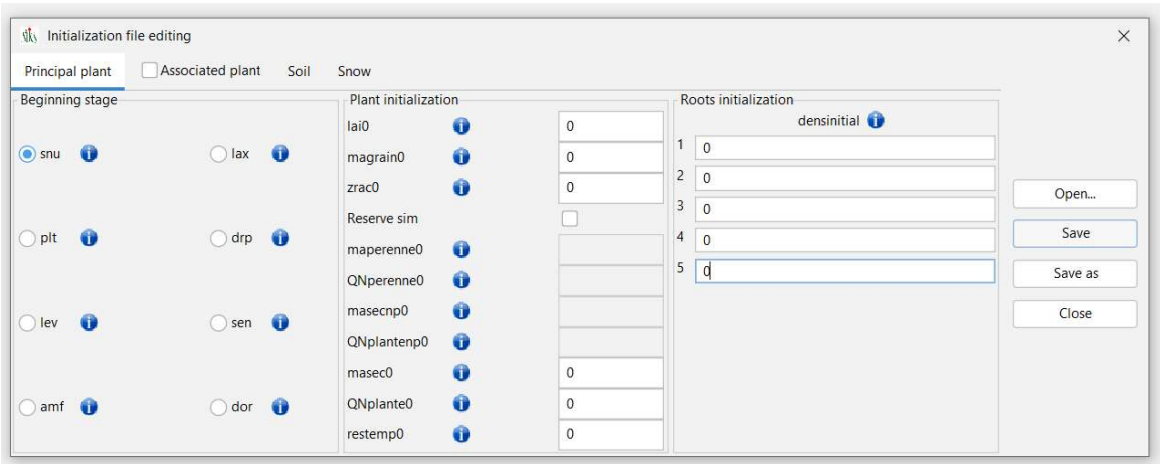
2. In JavaSTICS:



3. Fill in the file thanks to information given in the tab Ini of the inputs_stics_PW1 excel file: **/!\ be careful to put dots to separate the decimals!**

For the plant part, put 0 in the boxes, indeed the simulation begins in bare soil (snu).

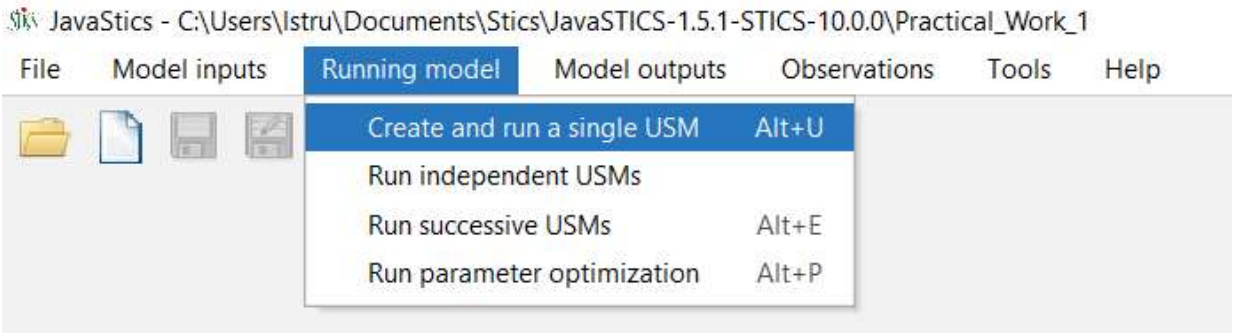
/!\ Do not let an empty cell !



4. In the "Soil" tab, a value of 0 means 0 for NH4 and NO3 content, but is interpreted as field capacity for the soil water content **/!\ Do not let an empty cell !**
5. SAVE AS Corn_ini.xml

V. Create a Unit of SiMulation: USM

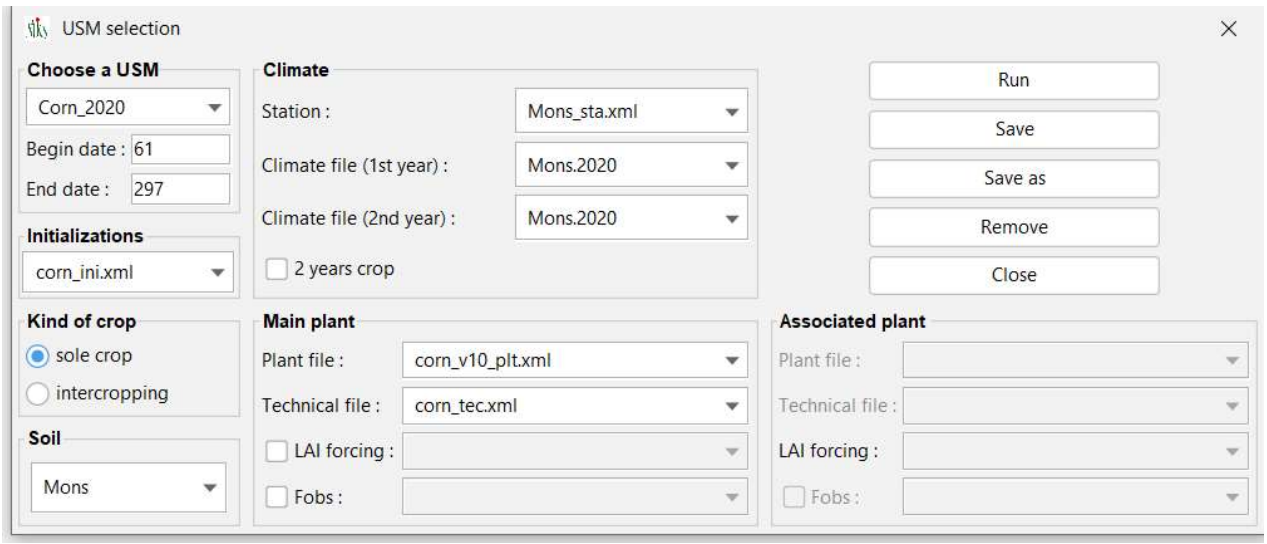
1. In JavaSTICS:



2. Create a USM with your input files from the default USM:

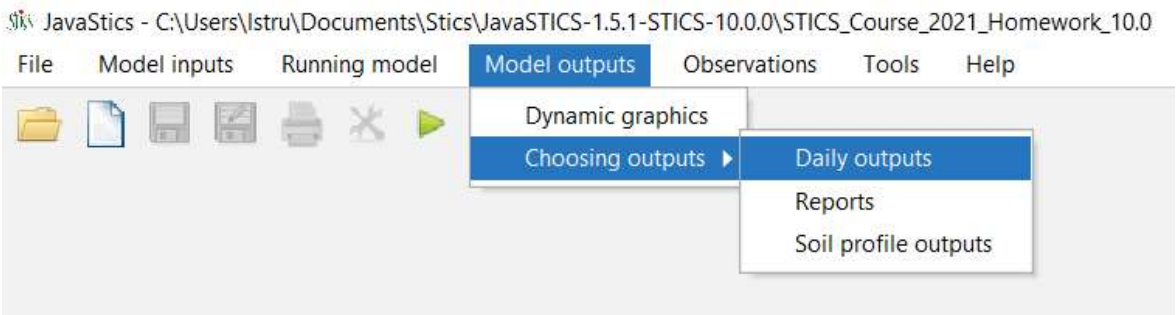
See tab "USMs" in the inputs_STICS_PW1 excel file

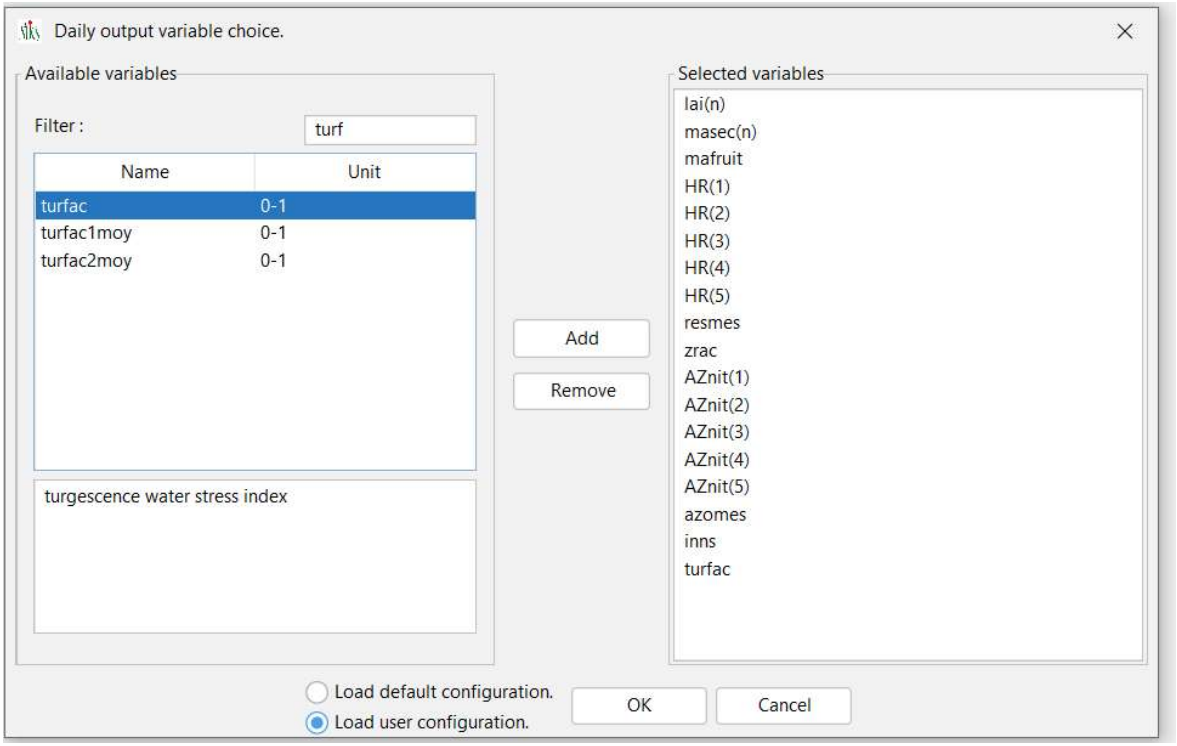
Change Begin date and End date
Select the good initialization file
We are in sole crop (nbplantes = 1)
Select the good soil
Select the good station file and associated climatic files: **/!\ don't forget to check the 2 years crop box (culturean = 2) if your simulation is on two years**
Select the good plant file
Select the technical file
We have no file of lai observation to use the lai forcing option (flai_1 = null)
Select the observation file
Save as Corn_2020



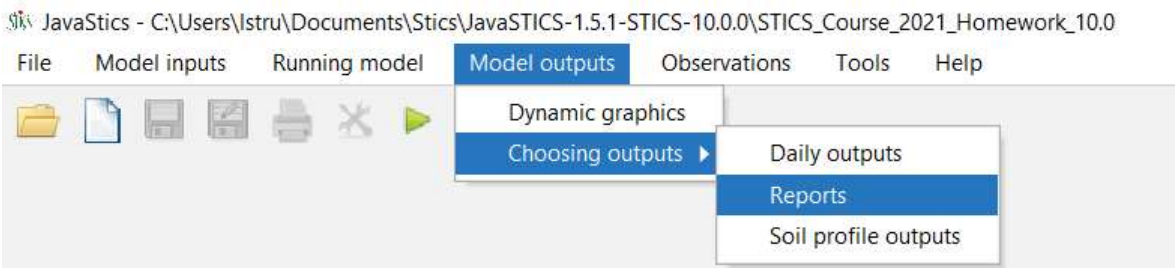
VI. Select the output data and run the USM

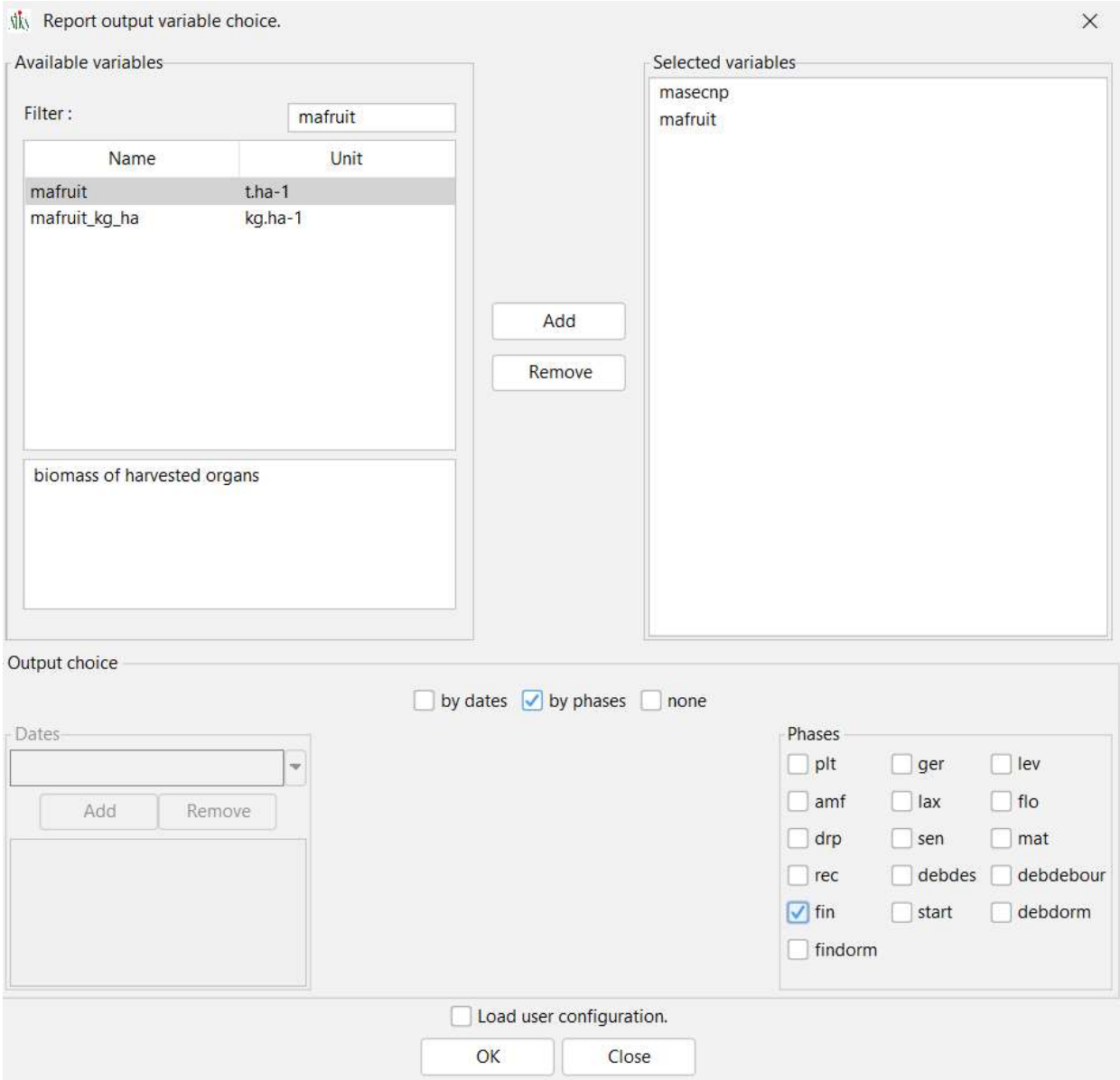
1. Selection of daily output variables in JavaSTICS



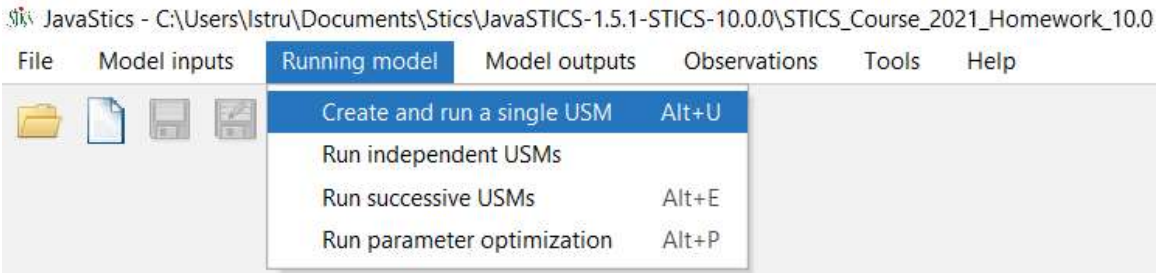


2. Selection of output variables at different dates or stages



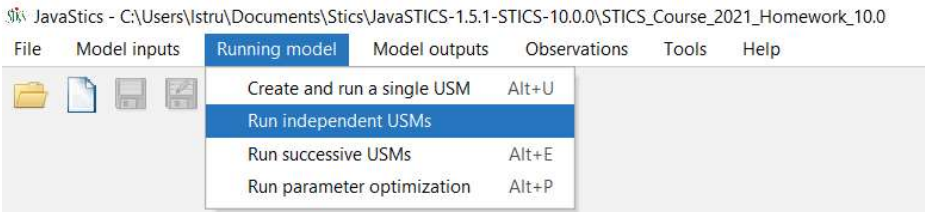


3. Run the simulation:
   - ✓ In JavaSTICS:



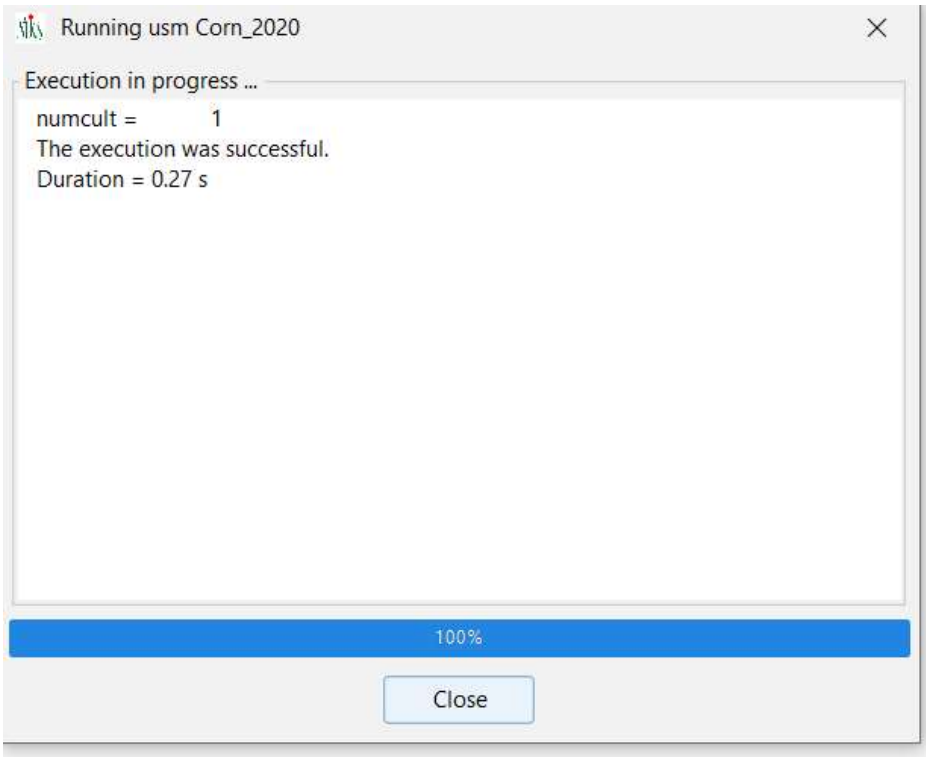
   - ✓ Select your USM
   - ✓ Run

Or:



- ✓ Select your USM
- ✓ Add
- ✓ Ok

You should have this message,



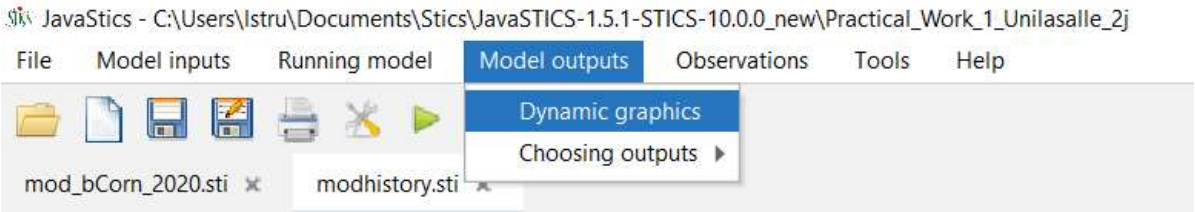
if not let me know! You did a mistake but that doesn't matter because you are here to learn!

VII. Have a look on the model outputs

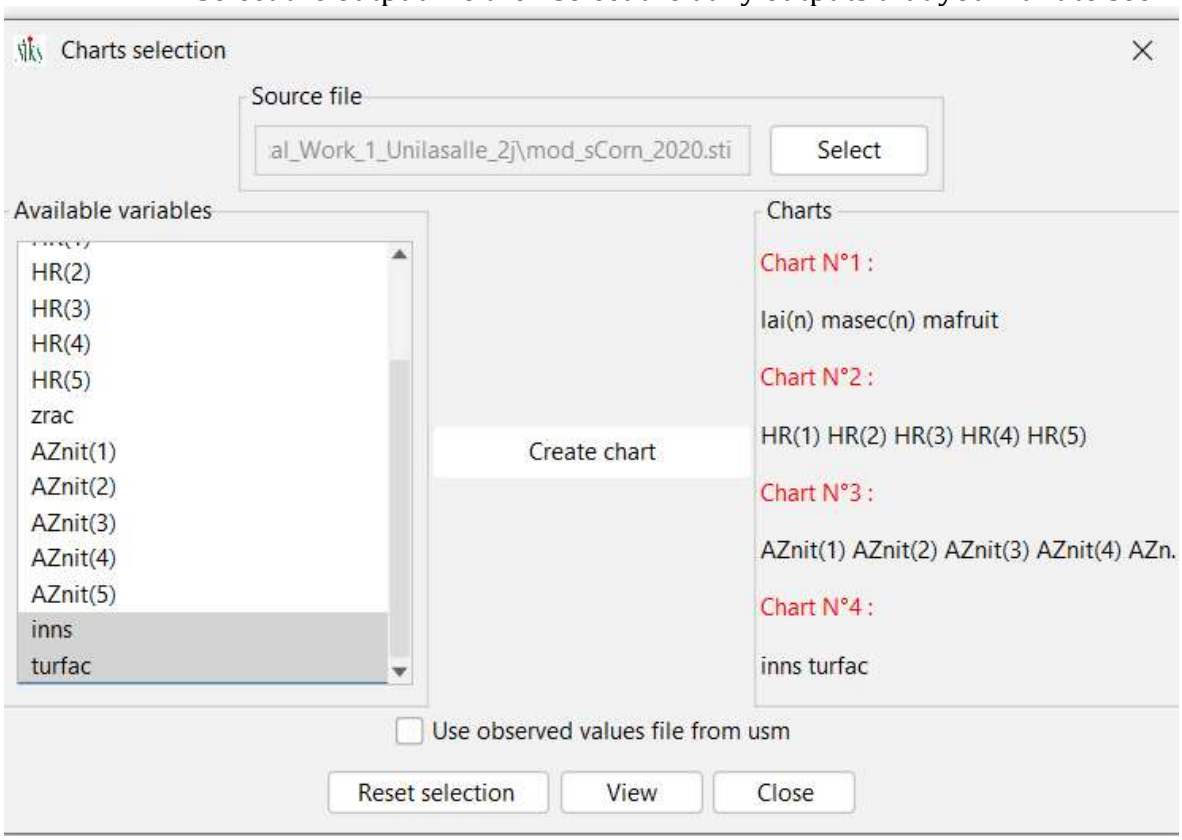
At the end of the simulation the model opens 2 files in the JavaSTICS interface:
modhistory.sti:
This file summarizes the parameters used for the simulation. If you have a problem during the simulation this is the only file created and it could say you why the simulation doesn't run!

mod_bCorn_2020.sti:
This file gives you the main information about the simulation results and balance of the simulation.

Graphics:
Do a graphic with the JavaSTICS interface.



1. Select the output file then select the daily outputs that you want to see



2. Click on "View"

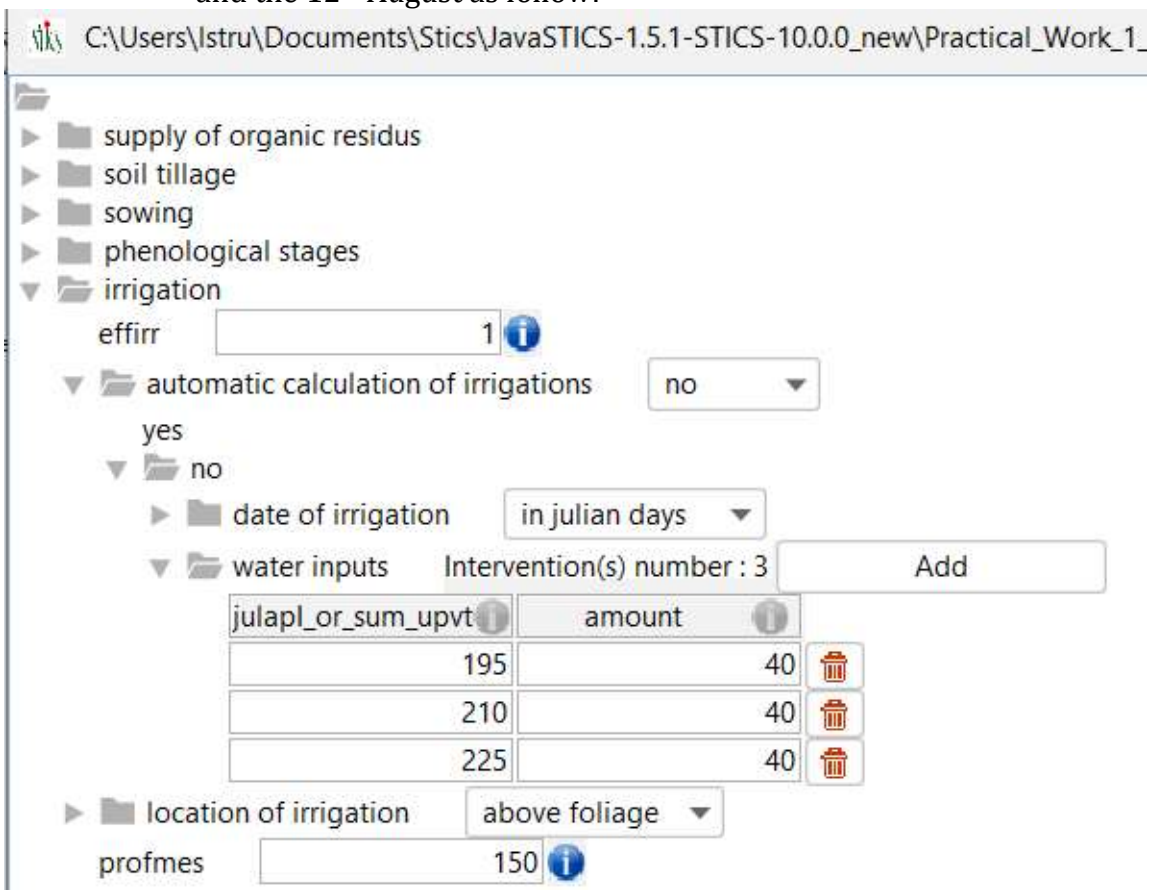
The files corresponding to the balance file, the daily outputs and to the report are stored in the working directory Practical_Work_1:
mod_bCorn_2020.sti: balance file
mod_sCorn_2020.sti: daily outputs
mod_rapport.sti: report

To open the file with excel, you just have to modify the file extension .sti by .csv

VIII. We are going to modify the crop management file to improve the biomass production

We saw that hydric stress was important during summer 2020. In order to improve the yield of the crop, we will add irrigation during this period.

1. Select the file called corn_tec.xml
2. Click on "New File"
3. Call the new file "corn_Irr"
4. Open the new file and add irrigation of 40 mm the 13th July, the 28th July and the 12th August as follow:



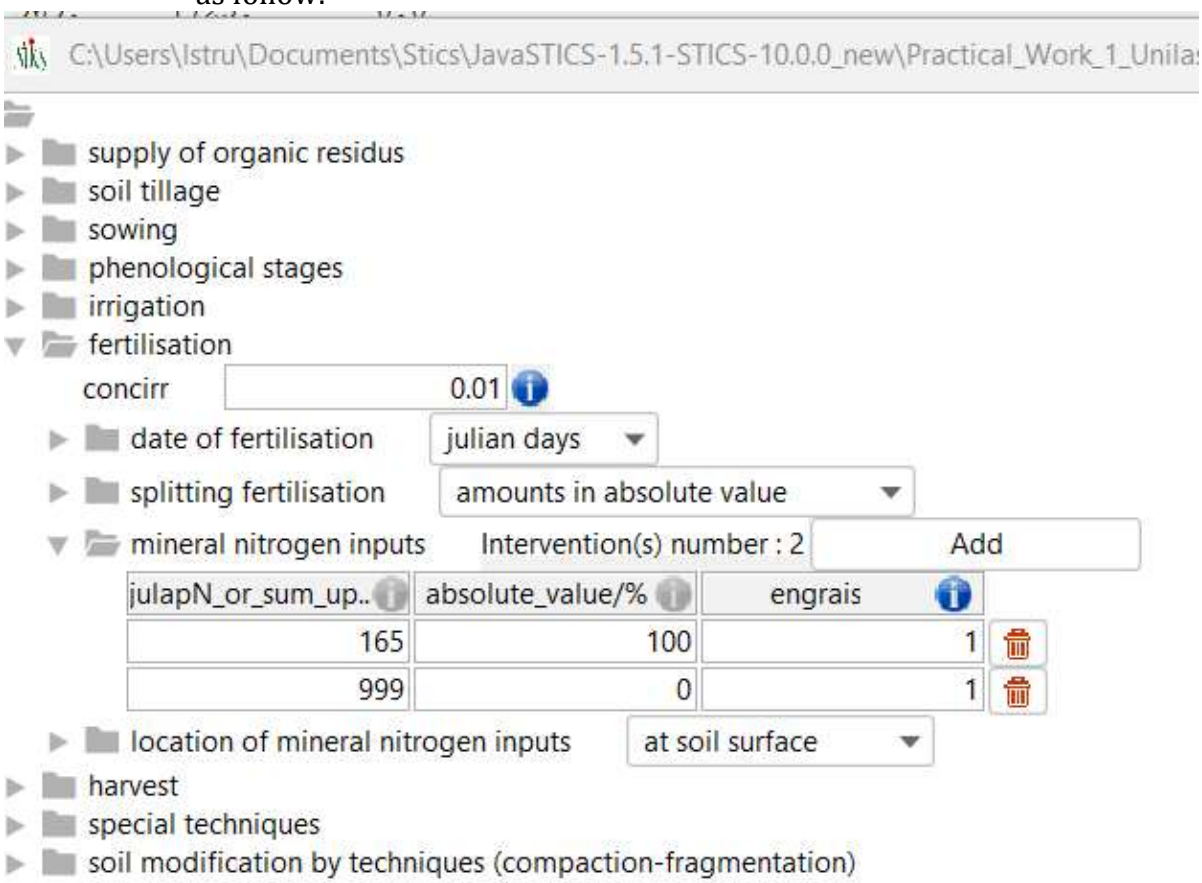
5. SAVE !

6. Select the USM called Corn_2020 and click and save as and called the new USM Corn_Irr_2020
7. Select the good technical file called “corn_Irr_tec.xml”
8. SAVE !
9. RUN

Let us have a look on the outputs.

We can see that we avoid a part of the hydric stress which led to an increase of biomass production and yield. Yield was enhanced thanks to an increase of grains number. However, we still have a low NNI (=lower than 1) and so the crop experienced a nitrogen stress.

1. Select the file called corn_Irr_tec.xml
2. Click on “New File”
3. Call the new file “corn_Irr_Fert”
4. Open the new file and add fertilization of ammonium nitrate the 13th June as follow:



5. SAVE !
6. Select the USM called Corn_2020 and click and save as and called the new USM Corn_Irr_Fert_2020
7. Select the good technical file called “corn_Irr_Fert_tec.xml”
8. SAVE !
9. RUN

Let us have a look on the outputs.

Here we used crop management to solve our problem. However, crop selection can also be a solution notably if we consider that water is not an inexhaustible resource. We could also say the same thing about mineral fertilizers but let us focus on water.
In order to have a crop that is more tolerant to water scarcity, what could we improve?

This is the end! Good job!